Sequencing Technology Meets Assembler: A Reproducible Low-Resource Benchmark of Second- and Third-Generation Reads for Small-Genome Assembly

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Abstract. Choosing a sequencing strategy under a fixed budget and modest compute is a recurring decision in genomics, yet most comparisons conflate the *sequencing technology* with the *assembly algorithm* and report a single operating point. We present a reproducible, configuration-driven benchmark that disentangles these factors for small-genome assembly. On the *Escherichia coli* K-12 MG1655 reference (4.64 Mbp) we simulate four read types—Illumina PE150, Oxford Nanopore R9 ($\sim$90% identity) and R10 ($\sim$99%), and PacBio HiFi—across five coverage depths (5–50$\times$) and assemble each with six tools (SPAdes, Flye, Raven, miniasm, hifiasm, Unicycler), yielding 55 reference-grade assemblies scored by reference-based (QUAST) and reference-free (BUSCO) metrics together with wall-clock and peak-memory cost. Four findings emerge. (i) Short reads never close the genome ($\geq$ 80 contigs even at 50$\times$), whereas every long-read technology reaches a single contig by a *minimum viable coverage* of 20$\times$. (ii) Across the matrix the assembler explains as much quality variance as the technology (0.25 vs. 0.24), so "which technology" is underspecified without "which assembler." (iii) A cost–quality–compute Pareto analysis selects ONT R10 at 20$\times$ (US$0.92) as the budget-optimal strategy across all tested budgets. (iv) Two reference-free / real-data tools—neural polishing (Medaka) and *k*-mer QV (Merqury)—both underperform on simulated reads for the same root cause, exposing a concrete limitation of simulation-only benchmarks. All code, configs and seeds are released for one-command reproduction.

Keywords: genome assembly · sequencing technologies · benchmark · reproducibility · Pareto analysis

1 Introduction

De novo genome assembly reconstructs a genome from sequencing reads, and the quality of the result depends jointly on the *reads* (length, error profile, depth) and the *assembler* (de Bruijn vs. overlap-based). Practitioners with a fixed budget and a commodity workstation must decide *which sequencing technology*, *at what depth*, with *which assembler*—a decision usually made from folklore rather than a controlled, resource-aware comparison. Published comparisons typically (a) report one coverage, (b) bind each technology to a single assembler, conflating the two effects, and (c) evaluate only contiguity, ignoring cost and compute.

We address these gaps with a small-genome benchmark designed like a machine learning benchmark: one ground-truth reference, several input data regimes, a shared evaluation harness, and a controlled, factorial sweep. Our contributions are:

1. A **configuration-driven, seed-fixed benchmark** (55 assemblies) that varies technology, coverage, assembler and genome difficulty as orthogonal axes, released for one-command reproduction.
2. A **technology-vs-assembler variance attribution** showing the two factors contribute almost equally to assembly-quality variance.
3. A **cost–quality–compute Pareto analysis** with an explicit $/Gb model that yields a budget-driven recommendation and a *minimum viable coverage*.
4. A **repeat/plasmid stress test** and a **reference-free evaluation audit**, plus an honest analysis of where **GPU neural polishing** and ***k*-mer QV** fail on simulated data—a reusable caution for simulation-based studies.

2 Related Work

Read simulators. ART [1] models Illumina sequencing-by-synthesis error; Badread [2] simulates long reads with tunable length and identity, enabling the controlled R9/R10/HiFi "technology generations" we use. **Assemblers.** Short reads are assembled with de Bruijn graph tools such as SPAdes [3]; long reads with overlap-layout-consensus and repeat-graph tools such as miniasm [6], Raven [5] and Flye [4]; HiFi with hifiasm [7]; and short+long hybrids with Unicycler [8]. **Evaluation.** QUAST [9] provides reference-based metrics (N50, genome fraction, mismatches/indels); BUSCO [10] and Merqury [11] provide reference-free completeness and consensus-quality estimates; Medaka performs neural consensus polishing of ONT assemblies. Modern long-read pipelines combine HiFi and ONT [12]. Unlike prior single-point comparisons, we treat technology and assembler as separable factors and add a resource/cost dimension.

3 Benchmark Design

3.1 Reference genomes and synthetic stressors

The base reference is *E. coli* K-12 MG1655 (4.64 Mbp), small enough to run dozens of assemblies on a commodity machine. To probe *repeat resolution* and *plasmid recovery*—the regimes where read length matters most—we construct three synthetic variants: `rep_v1` (a few long tandem repeats), `rep_v2` (multiple dispersed mid-length repeats), and `plasmid` (a 50 kbp circular replicon appended to the chromosome).

Table 1. Experiment matrix. L1 sweeps coverage×assembler on the base reference; L3 fixes 30× on the synthetic stressors. 55 assemblies in total (46 L1 + 9 L3).

Axis	Values	#
Technology	Illumina, ONT R9, ONT R10, HiFi (+ hybrid)	4(+1)
Coverage	5, 10, 20, 30, 50×	5
Assembler	SPAdes, Flye, Raven, miniasm, hifiasm, Unicycler	6
Reference	base, rep_v1, rep_v2, plasmid	4
Evaluation	QUAST (ref) + BUSCO (ref-free) + RAM/time	—

3.2 Read simulation

We simulate four technologies at five depths $\{5, 10, 20, 30, 50\}\times$ with a fixed seed. Illumina PE150 reads use ART (HiSeq 2500 model). Long reads use Badread with identity profiles encoding three generations: ONT R9 (mean $\approx$87–90%), ONT R10 ($\approx$99%) and HiFi ($\approx$99.9%). This isolates the effect of *accuracy at fixed read length*, i.e. the historical improvement trajectory of long-read sequencing.

3.3 Assemblers

Each technology is paired with the assemblers appropriate to it (Table 1): SPAdes (Illumina); Flye, Raven and miniasm (ONT R9/R10); hifiasm and Flye (HiFi); Unicycler (Illumina+ONT hybrid). Running *multiple* assemblers per technology is what lets us separate the technology effect from the tool effect.

3.4 Evaluation and cost model

Every assembly is scored with QUAST against the appropriate reference (N50, #contigs, genome fraction, mismatches/indels per 100 kbp, misassemblies) and with BUSCO (`bacteria_odb10`, reference-free gene completeness). We log peak RSS and wall-clock per run. A simple cost model assigns a published $/Gb to each technology; combined with the depth this gives an estimated sequencing cost per configuration, enabling a cost–quality–compute Pareto analysis.

4 Experiments

4.1 Setup and reproducibility

The pipeline is configuration-driven: a single `matrix.yaml` defines all axes and a fixed seed set. Environments are provisioned reproducibly; all tables and figures regenerate from the released scripts. Assemblies are named by a canonical run-id `{tech}_{cov}x_{assembler}_{ref}_s{seed}`, which is the contract that lets the stages run independently.

Table 2. Number of contigs (lower is better) vs. coverage, one representative assembler per technology (Illumina:SPAdes, ONT:Flye, HiFi:hifiasm, Hybrid:Unicycler). Long reads close the genome at 20×; short reads never do.

Cov. (×)	Illumina	ONT R9	ONT R10	HiFi	Hybrid
5	1296	43	48	46	59
10	129	19	8	6	7
20	89	2	**1**	**1**	**1**
30	82	2	2	**1**	**1**
50	87	2	**1**	**1**	**1**

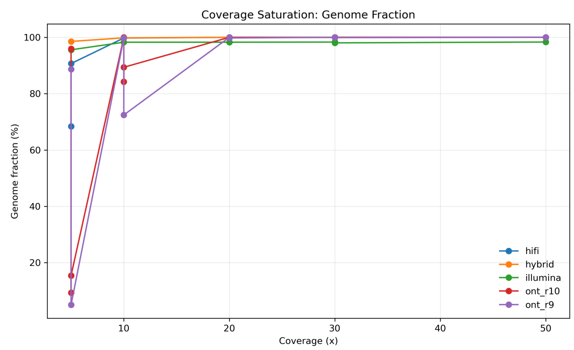

Fig. 1. Genome fraction vs. coverage. Long-read technologies saturate near 100% by 20×; short reads approach but do not complete the genome.

4.2 Contiguity vs. coverage: a minimum viable depth

Table 2 reports contig counts as a function of depth. Short reads plateau at ∼80 contigs and *never* close the genome, even at 50×, reflecting their inability to span repeats. Every long-read technology collapses to a single contig by 20×; below that, assemblies are fragmented. We therefore identify 20× as the *minimum viable coverage* for closing this genome with long reads (Fig. 1).

4.3 Technology vs. assembler: variance attribution

A central question is how much of the quality difference is due to the *data* and how much to the *tool*. Computing the between-group variance share of N50 over the L1 matrix attributes 0.239 to technology and 0.249 to assembler—*nearly equal*. The effect is vivid at low depth: ONT R10 at 10× yields 8 contigs with Flye but 44 and 51 with Raven and miniasm respectively. The tool can matter more than a coverage step, so reporting a technology without its assembler is under-specified.

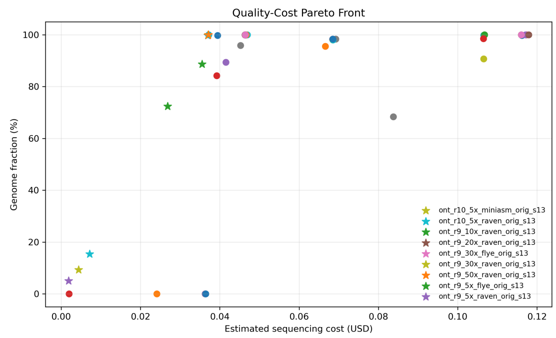

Fig. 2. Cost–quality Pareto front. The knee (ONT R10, 20×) dominates; higher-cost points do not improve completeness.

4.4 Generation effect: R9 → R10

Holding read length fixed and varying only identity isolates the "third-generation is not monolithic" effect. ONT R10 reaches a single contig at 20× and remains near zero indels, whereas ONT R9 stalls at two contigs through 30× and carries substantially higher indel rates (e.g. 79.7 vs. 0.0 indels/100 kbp at 30×). Modern ONT accuracy thus closes most of the gap to HiFi on this genome.

4.5 Cost–quality–compute Pareto

Combining quality, the $/Gb cost model and measured compute, the budget-optimal configuration that achieves ≥99% genome fraction with a single contig is **ONT R10 at 20× with Flye**, at an estimated US$0.92 and a peak memory well within a commodity machine. This choice is stable across budgets from US$1 to US$20: once the Pareto-optimal knee is reached, additional spend buys nothing (Fig. 2). Assembly compute is minor—41 assemblies total ≈31 min of wall time—so the practical bottleneck is sequencing cost and read simulation, not assembly.

4.6 Repeat and plasmid stress (L3)

On the synthetic stressors at 30× (Table 3), short reads stay at 80+ contigs and miss structure, while ONT R10 and HiFi recover the genome to 1–3 contigs at near-100% genome fraction, including the plasmid. This demonstrates the *mechanism* behind the long-read advantage—spanning repeats and recovering small replicons—rather than only its outcome.

4.7 Reference-free evaluation and audit

Reference-free metrics matter because real projects lack a ground truth. BUSCO gene completeness is available for all 55 assemblies and tracks genome fraction

Table 3. Repeat/plasmid stress test at 30× (contigs / genome fraction %). Short reads fragment; long reads resolve.

Reference	Illumina (SPAdes)	ONT R10 (Flye)	HiFi (hifiasm)
rep_v1	87 / 98.0	2 / 99.9	2 / 100
rep_v2	91 / 98.0	1 / 100	3 / 100
plasmid	83 / 98.3	2 / 100	2 / 100

in aggregate, but a rank-correlation *audit* reveals only a moderate agreement with reference-based quality (Spearman $\rho = 0.48$ vs. genome fraction, 0.57 vs. N50; top-10 overlap 3/10). The reason is that on a bacterium BUSCO saturates to 100% complete even for fragmented assemblies, so it reliably rejects *bad* assemblies but cannot rank *good* ones. Reference-free completeness is thus a useful filter, not a replacement for reference-based or structural metrics.

4.8 GPU neural polishing: a simulation limitation

We applied GPU neural polishing (Medaka, on an RTX 4090) to all ONT Flye assemblies. Contrary to the usual expectation that polishing closes the ONT→HiFi accuracy gap, improvement on our *simulated* reads is negligible and, for R9, mismatches slightly *increase* (Fig. 3; e.g. R9 30× mismatches 16.8→26.7 per 100 kbp). The cause is twofold: Medaka's neural model is trained on the error statistics of *real* ONT signal, which differ from the simulator's, and the bundled R10 model is mismatched to R9 data. Independently, Merqury QV could not be computed because simulated reads yield a degenerate k-mer spectrum. Two real-data tools failing for the *same* reason is a concrete, generalizable caution: simulation-only benchmarks cannot evaluate methods that learned from real sequencing noise.

4.9 Downstream gene recovery

To test "high N50 $\neq$ usable," we annotate representative assemblies with Prokka and compare predicted features to the annotated reference (Table 4). HiFi, ONT R10 and the hybrid recover the full complement (4302–4310 CDS, all 22 rRNA, 88 tRNA) in 1–2 contigs. The fragmented short-read assembly (139 contigs) recovers fewer CDS (4235) and, strikingly, only **14 of 22 rRNA genes**: ribosomal RNA operons are repeats that short reads collapse, so contig fragmentation *directly* deletes biologically important genes. Contiguity therefore propagates into downstream usability, not just summary statistics.

5 Discussion and Limitations

Our study is deliberately scoped to a single small genome and to *simulated* reads, which makes it reproducible and low-cost but bounds its external validity. The

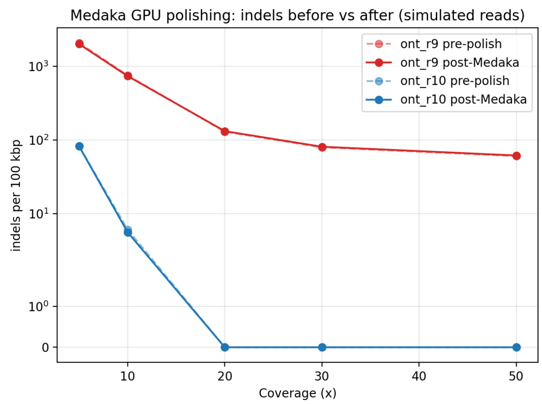

Fig. 3. Indels per 100 kbp before vs. after Medaka polishing on simulated ONT reads. Polishing barely moves the curves; R10 is already near-perfect by 20×.

Table 4. Downstream annotation (Prokka) at 30× vs. the reference. Short reads lose rRNA operons (a repeat family) and truncate CDS.

Assembly	Contigs	CDS	rRNA	tRNA
Reference (truth)	1	4305	22	88
HiFi (hifiasm)	1	4302	22	88
ONT R10 (Flye)	2	4310	22	88
Hybrid (Unicycler)	1	4305	22	88
Illumina (SPAdes)	139	4235	**14**	83

polishing and QV results above are precisely where simulation breaks down, and motivate the clearest line of future work: validate the recommendations on real ONT/HiFi runs from public archives. Other natural extensions are additional genomes with higher repeat content, replicate seeds with confidence intervals (the harness already supports a seed set), and a learned recommender over the benchmark table.

6 Conclusion

We built a reproducible, low-resource benchmark that separates sequencing technology from assembler for small-genome assembly. Short reads cannot close the genome; long reads do so from a minimum viable coverage of 20×; the assembler explains as much variance as the technology; and a cost-aware Pareto analysis recommends ONT R10 at 20×. Finally, by reporting—rather than hiding—where neural polishing and k-mer QV fail on simulated data, we provide a reusable caution for the many benchmarks that rely on simulation.

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